Assignment 5


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Question

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Implement the Stop-and-Wait and Stop-and-Wait ARQ (Automatic Repeat reQuest)
protocols. Simulate a communication channel with a frame loss probability of
70% and demonstrate the operation of both protocols. Verify and report the
observed frame loss rate and compare the retransmission behaviour of the two
protocols.
```

CODE

```python
import random

# ============================================================
# CONFIGURATION
# ============================================================

FRAME_LOSS_PROB = 0.7
TOTAL_FRAMES = 100


# ============================================================
# COMMUNICATION CHANNEL
# ============================================================

class Channel:

    def transmit(self, frame_id):
        """
        Simulates transmission of a frame.

        Returns True if transmission succeeds,
        otherwise returns False.
        """

        is_success = random.random() >= FRAME_LOSS_PROB

        if not is_success:
            print(f"{frame_id},", end="")
```

```
        return is_success


# ============================================================
# STOP-AND-WAIT PROTOCOL
# ============================================================

def stop_and_wait(ch):

    total_count = 0
    success_count = 0
    failed_count = 0

    print("\nFailed attempts (Frame IDs): ", end="")

    for frame_id in range(1, TOTAL_FRAMES + 1):

        total_count += 1

        if ch.transmit(frame_id):
            success_count += 1
        else:
            failed_count += 1

    print("\n")

    result = {
        "total transmitted": total_count,
        "successful": success_count,
        "failed": failed_count,
        "re-transmitted": 0,
        "loss-rate": (failed_count * 100) / total_count,
        "delivery-rate": (success_count * 100) / total_count
    }

    return result


# ============================================================
# STOP-AND-WAIT ARQ PROTOCOL
# ============================================================

def stop_and_wait_arq(ch):

    total_count = 0
    success_count = 0
    failed_count = 0
    retransmitted = 0

    print("\nFailed attempts (Frame IDs): ", end="")

    for frame_id in range(1, TOTAL_FRAMES + 1):

        while True:

            total_count += 1

            if ch.transmit(frame_id):
                success_count += 1
```

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                break

            else:
                retransmitted += 1

    print("\n")

    result = {
        "total transmitted": total_count,
        "successful": success_count,
        "failed": failed_count,
        "re-transmitted": retransmitted,
        "loss-rate": (retransmitted * 100) / total_count,
        "delivery-rate": (success_count * 100) / total_count
    }

    return result


# ============================================================
# DISPLAY RESULTS
# ============================================================

def display_results(result):

    for key, value in result.items():

        if "rate" in key:
            print(f"{key} : {value:.2f}%")
        else:
            print(f"{key} : {value}")


# ============================================================
# MAIN
# ============================================================

if __name__ == "__main__":

    ch = Channel()

    print("---------------- Stop and Wait ----------------")

    snw_result = stop_and_wait(ch)
    display_results(snw_result)

    print("\n-------------- Stop and Wait ARQ --------------")

    snw_arq_result = stop_and_wait_arq(ch)
    display_results(snw_arq_result)
```

OUTPUT

```
---------- Stop and Wait ----------

Failed attempts (Frame IDs): 1,2,3,4,6,8,10,11,13,15,16,17,18,19,20,24,26,27,29,30,31,32,33,34,36,37,40,41,42,43,44,45,49,50,51,
53,54,55,56,57,59,60,61,63,64,66,67,68,70,72,73,74,75,76,77,80,81,82,85,86,87,90,91,92,93,96,97,98,99,

total transmitted : 100
successful : 31
failed : 69
re-transmitted : 0
loss-rate : 69.00%
delivery-rate : 31.00%

---------- Stop and Wait ARQ ----------

Failed attempts (Frame IDs): 2,2,2,2,2,2,2,2,2,2,3,3,4,4,5,5,5,6,7,7,8,11,11,12,12,12,13,13,15,15,18,18,20,20,20,21,21,21,23,2
3,23,24,24,24,25,26,26,27,27,27,27,27,27,27,27,27,28,29,29,29,29,29,29,29,29,29,31,31,31,32,32,32,32,32,33,33,33,33,33,33,33,
33,34,34,34,36,38,38,38,38,41,45,45,45,45,45,45,45,46,47,48,48,48,50,50,50,50,50,50,50,50,53,54,55,55,56,56,57,57,57,57,60
,60,62,63,66,66,66,66,66,66,69,69,70,72,73,73,74,75,75,77,77,77,78,78,78,78,78,78,78,78,79,79,79,79,81,81,81,81,81,81,81,82,8
2,82,83,84,84,84,84,84,84,84,84,84,84,84,84,84,84,85,87,87,87,87,87,87,88,88,89,90,91,91,91,91,91,92,95,96,96,96,96,96,96,98,98,
100,100,100,100,100,

total transmitted : 316
successful : 100
failed : 0
re-transmitted : 216
loss-rate : 68.35%
delivery-rate : 31.65%
```